Checkpoint C | Version 1 | Solutions

*Each skill earns **Pass** or **Not Yet**. Some questions contain multiple skills. Grade each skill on its own when it's part of a full question. A **Pass** is given for work showing the core skill. A **Not Yet** is given when there's not sufficient evidence of the core skill. Evidence anywhere on the submission is acceptable but evidence in the question area will be used for the initial grading. Students may request a regrade if evidence exists outside the original grading area and was not included in the original grading. Do not infer a method that the work does not support.*

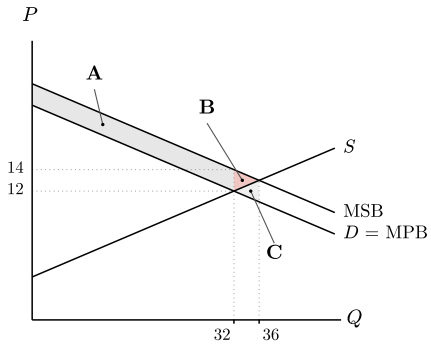
Lovely Lighting from Lumos Lanterns

Many Hogsmeade homeowners hang Lumos Lanterns outside their doors. In addition to providing lighting for the homeowner, each lantern also lights the lane for the neighbors, an external benefit of 2 coins per lantern. Prices are in coins and quantities in lanterns. The demand and supply equations are:

$$D: P_b = 20 - \frac{1}{4}Q_d \qquad S: P_s = 4 + \frac{1}{4}Q_s$$

C1.2 | Market Failure

The graph shows the market for lanterns: marginal private benefit (MPB), marginal social benefit (MSB), and supply (S). The lettered regions are for the questions.



Market Quantity ($\hat{Q}$): _____

Efficient Quantity (Q^*): _____

Deadweight Loss (*letter*): _____

Deadweight Loss (*area*): _____

For the Last Lantern Traded: MSB: _____ / MC: _____

The Market Trades: ☐ too many ☐ too few lanterns

Answers: market quantity **32 lanterns**; efficient quantity **36 lanterns**; DWL region **B**; DWL area **4 coins**; last lantern traded: **MSB = 14**, **MC = 12**; **too few** lanterns.

The external benefit shifts MPB up by 2 at every quantity:

$$\text{MSB} = 20 - \frac{Q}{4} + 2 = 22 - \frac{Q}{4}.$$

The market trades where private benefit equals cost, while efficiency counts the neighbors' benefit:

$$20 - \frac{Q}{4} = 4 + \frac{Q}{4} \Rightarrow \hat{Q} = 32, \quad \hat{P} = 12$$
$$22 - \frac{Q}{4} = 4 + \frac{Q}{4} \Rightarrow Q^* = 36.$$

At the market quantity, $\text{MSB} = 22 - \frac{32}{4} = 14$ and $\text{MC} = 4 + \frac{32}{4} = 12$. Additional lanterns between 32 and 36 would provide more social benefit than they cost, but private buyers do not purchase them. That lost surplus is region B:

$$\text{DWL} = \frac{1}{2} \times 2 \times (36 - 32) = 4 \text{ coins}.$$

A is the external benefit of lanterns already traded. C is the gap between MC and MPB for the additional socially worthwhile trades, not the loss to society.

Pass (C1.2): Identify the market and efficient quantities, identify B and calculate its area, and compare MSB with MC at the market quantity to explain why too few lanterns are traded. No separate MSB-gap answer is required.

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C2.2 | A Subsidy for Lanterns

To brighten the village, the Ministry pays sellers a subsidy of 4 coins per Lumos Lantern sold. For this question, ignore the external benefit: demand is the marginal benefit (MB) of a lantern.

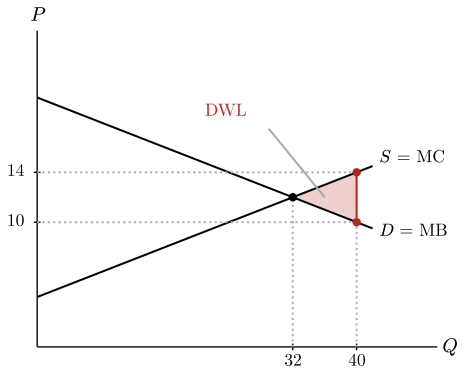
Buyer's Price (P_b): _____

Seller's Price (P_s): _____

Subsidy Quantity (Q^{Sub}): _____

Deadweight Loss (DWL): _____

Last Lantern Traded, MC vs. MB: ☐ above ☐ below



Answers: buyer's price **10 coins**; seller's price **14 coins**; subsidy quantity **40 lanterns**; DWL **16 coins**; last lantern traded: MC is **above** MB.

For this question the external benefit is ignored, so the unsubsidized market is efficient at $Q = 32$, $P = 12$. The subsidy makes the seller's price 4 coins higher than the buyer's:

$$P_s = P_b + 4 \quad \Rightarrow \quad 4 + \frac{Q}{4} = (20 - \frac{Q}{4}) + 4 \quad \Rightarrow \quad Q^{Sub} = 40.$$

$$P_b = 20 - \frac{40}{4} = 10, \quad P_s = 4 + \frac{40}{4} = 14.$$

Check: $14 - 10 = 4$. At $Q = 40$, MC is 14 while MB is 10. Every added trade between 32 and 40 costs more than its private benefit. The resulting loss is the triangle between supply and demand:

$$\text{DWL} = \frac{1}{2} \times 4 \times (40 - 32) = 16 \text{ coins}.$$

The relevant comparison is MC versus MB, not MSB; the efficient baseline for this question is 32, not the 36 from C1.2.

Pass (C2.2): Use the subsidy wedge to find quantity and both prices, identify MC above MB for the last trade, and calculate the 16-coin DWL using the no-externality baseline.

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Cauldrons For Days

The Standard Cauldron is a magical tool used widely in the making of various potions with many buyers and many sellers. The demand and supply curves for the Standard Cauldron can be represented by the following relationships, with prices in coins, and quantities in cauldrons.

$$D : P_b = 120 - Q_d \qquad S : P_s = 30 + \frac{1}{2}Q_s$$

The metal for each cauldron is smelted in foundries outside Hogsmeade, and the fumes drift over the village. The fumes cost residents 15 coins per cauldron made, a cost neither buyers nor sellers pay.

C1.1 | Market Failure

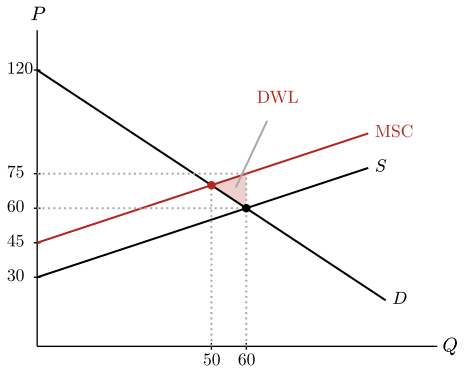
Start by using a graph and algebra to describe the market failure. Graph demand, supply, and MSC, and shade the deadweight loss.

Market Quantity ($\hat{Q}$): _____

Efficient Quantity (Q^*): _____

Deadweight Loss (DWL): _____

The Market Trades: ☐ too much ☐ too little



Answers: market quantity **60 cauldrons**; efficient quantity **50 cauldrons**; DWL **75 coins**; **too much**. The graph must show MSC $= 45 + \frac{Q}{2}$ and shade the loss between MSC and demand from 50 to 60.

Add the external cost to private marginal cost:

$$\text{MSC} = 30 + \frac{Q}{2} + 15 = 45 + \frac{Q}{2}.$$

Market equilibrium and efficiency respectively require:

$$120 - Q = 30 + \frac{Q}{2} \Rightarrow \hat{Q} = 60, \quad \hat{P} = 60$$

$$120 - Q = 45 + \frac{Q}{2} \Rightarrow Q^* = 50.$$

At $Q = 60$, MSC $= 75$ but MB $= 60$. Trades beyond 50 cost society more than they benefit buyers. The DWL triangle has corners $(50, 70)$, $(60, 75)$, and $(60, 60)$:

$$\text{DWL} = \frac{1}{2} \times 15 \times (60 - 50) = 75 \text{ coins}.$$

Pass (C1.1): Construct MSC by adding the external cost, find market and efficient quantities, correctly shade and calculate DWL, and identify overproduction. The student sheet asks for MSC on the graph, not a separate equation blank.

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C2.1 | A Tax for Revenue

Needing revenue, and not yet knowing the size of the externality, the Ministry imposes a tax of 9 coins per cauldron. For this question, ignore the external cost.

Buyer's Price (P_b): _____

Seller's Price (P_s): _____

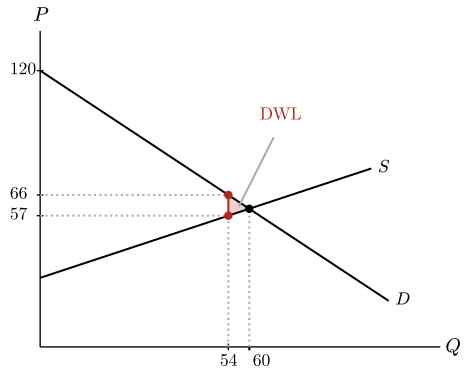
Post-Tax Quantity (Q^{Tax}): _____

Tax Incidence on Buyers: _____

Tax Incidence on Sellers: _____

Deadweight Loss (DWL): _____

Who Bears More of the Tax? ☐ buyers ☐ sellers ☐ equal



Answers: buyer's price **66 coins**; seller's price **57 coins**; post-tax quantity **54 cauldrons**; tax incidence **6 coins per cauldron on buyers** and **3 on sellers**; DWL **27 coins**; **buyers** bear more.

The tax makes the buyer's price 9 coins higher than the seller's:

$$P_b = P_s + 9 \quad \Rightarrow \quad 120 - Q = 30 + \frac{Q}{2} + 9 \quad \Rightarrow \quad Q^{Tax} = 54.$$

$$P_b = 120 - 54 = 66, \quad P_s = 30 + \frac{54}{2} = 57.$$

Check: $66 - 57 = 9$. Against the untaxed price of 60, buyers pay $66 - 60 = 6$ more and sellers receive $60 - 57 = 3$ less. Buyers bear more of the tax.

Ignoring the external cost, the efficient quantity is the untaxed quantity of 60. The tax prevents six mutually beneficial trades, creating a triangle between demand and supply:

$$\text{DWL} = \frac{1}{2} \times 9 \times (60 - 54) = 27 \text{ coins}.$$

Use the no-externality baseline of 60 here, not the socially efficient 50 from C1.1.

Pass (C2.1): Use the tax wedge to find quantity and both prices, measure both burdens against the untaxed price and identify buyers as bearing more, and calculate DWL with the external cost ignored.

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C3.1 | Designing the Policy

The Ministry of Magic has hired you to design a policy that corrects the market failure caused by the negative externality of cauldron production. Propose the policy you recommend and how it would impact the market for cauldrons and their users.

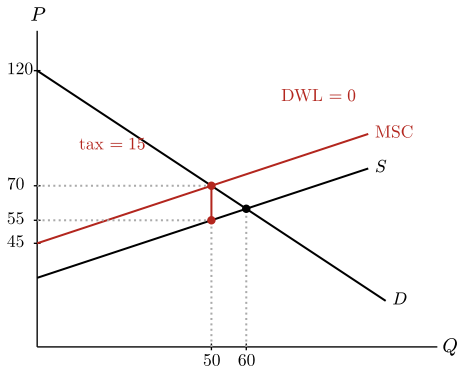
Policy Type: _____

Policy Size: _____

Post-Policy Quantity (Q^τ): _____

Post-Policy Deadweight Loss: _____

Post-Policy Quantity vs. Efficient: ☐ above ☐ at ☐ below



Answers: a per-unit **tax** of **15 coins per cauldron**; post-policy quantity **50 cauldrons**; post-policy DWL **0 coins**; quantity is **at** the efficient level.

Now count the 15-coin external cost again. Set the tax equal to that cost, replacing the earlier 9-coin revenue tax. Supply plus the tax coincides with MSC:

$$P_b = P_s + 15 \quad \Rightarrow \quad 120 - Q = 30 + \frac{Q}{2} + 15 \quad \Rightarrow \quad Q^\tau = 50.$$

The tax removes the privately attractive but socially costly trades between 50 and 60. At 50, MB equals MSC, so the policy reaches the efficient quantity and leaves no deadweight loss.

Cauldron users pay $120 - 50 = 70$ coins and sellers keep $30 + \frac{50}{2} = 55$ coins. Relative to the unregulated market, users buy 10 fewer cauldrons and the fumes harm residents less. Policy prices are explanatory, not additional required answer blanks.

Pass (C3.1): Choose a tax equal to the 15-coin external cost, calculate the resulting quantity, identify it as efficient, and give zero remaining DWL. The comparison concerns the corrective policy, not the earlier revenue tax. Assess this skill independently of arithmetic errors in C1.1 or C2.1.